Carisurg

Assignment 22

Interim Documentation

Due Date: August 5th, 2026.

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0.0. Deployment Context

This package covers two candidate deployments of the triage classification model (logistic regression baseline, class_weight='balanced', top-10 features, ESI-1 recall 0.750 on the Yale EMMLC dataset) built in Weeks 0–7:

1. **Setting A** — ED Triage Desk: a screen-based, nurse-facing HCI implementation.
2. **Setting B** — Observation Unit: a kiosk/robot-assisted HRI implementation situated beside a vitals station.

Sections 1 and 2 follow canvases 01 (Problem space), 02 (Ethics), 03 (Guidelines), 04 (MVP), 05 (Environment), and 06 (Form) of the SoRoCo canvas set — the six canvases the CariSurg tutorials selected out of the ten in the full framework.

Per the tutorial's own guidance, the canvases were originally designed for HRI (robot) applications; fields such as "Robot's role" or "Appearance" are edited for Setting A where they do not map directly onto a screen-based interface, and this is flagged inline wherever it happens.

1.0. Co-Design Canvas

Setting A: ED Triage Desk (HCI)

1.1. Problem Space

What problem are you solving?

User			
Who are the users? Are there supporting users?	Group(s)	Characteristics	Needs
Primary Users	ED triage nurses (e.g. Sister Alleyne's team) — trained clinical staff, 12-hour shifts.	Time-pressured, constant task-switching, variable shift fatigue, trained in ESI acuity not ML confidence.	Classification actionable in seconds; visible override; no added data-entry burden.
Secondary Users	Shift supervisor; Clinical IT Lead (Martina Griffith); Integration Review Board.	Supervisory/oversight role; reviews logs periodically, not continuously.	Confidence overrides are logged/reviewable; evidence of safety before wider rollout.

What are your objectives?

Goal(s)		
What do the primary and secondary users want to accomplish?	Short-Term	Long-Term
Primary Users	Reduce inconsistency between presenting acuity and assigned triage category.	Fewer missed ESI-1 patients; a defensible, auditable triage record.
Secondary Users	Confidence overrides are logged and reviewable.	Evidence base for extending the model to other Mercer departments.

1.2 Ethical Considerations

The template's grid structure (Problem/Solution × User/Robot) is used as printed; category names (Automation Bias, Accountability, Data Governance) are chosen for this deployment in place of the template's own examples ("Physical safety", "Data security"), which the tutorial explicitly permits for non-robot settings.

Automation Bias		
	Problem	**Solution**
User	Nurses may defer to the model's classification rather than exercising independent clinical judgement, especially under fatigue.	Interface never presents the classification as a decision, only a flag; overriding requires an active, logged action.
Robot / System	The model's typically-correct output could be misread by the interface design as more certain than it is.	A confidence indicator is always shown alongside the classification; low-confidence outputs are visually distinct from high-confidence ones.

Accountability		
	Problem	**Solution**
User	If the nurse overrides the model's output and the patient later deteriorates, responsibility for the outcome is unclear.	Every override is timestamped, logged with a selected reason, and reviewable by the shift supervisor.
Robot / System	The model's recommendation could be treated as the "official" record rather than one input to it.	The system log stores the model's output and the human's decision as equally weighted, separately labelled fields — neither overwrites the other.

Data Governance		
	Problem	**Solution**
User	A patient in acute distress cannot meaningfully consent to a separate, model-specific data-use request at triage.	Model use is folded into Mercer's standard ED admission consent rather than added as a new friction point.
Robot / System	The Yale EMMLC dataset behind the model was not collected at Mercer, so it holds patient vitals data whose local applicability is not yet proven.	Data is used only for the immediate triage decision; no secondary use (e.g. retraining) without separate governance approval.

1.3 Guidelines

Field labels for Canvas 03 were not directly visible in the available tutorial screenshots; the questions below follow the tutorial's description of this canvas's purpose (interaction type, patient/clinician interaction, and ethics-to-design translation).

Appropriate interaction type	Passive decision support only — a classification and rationale surfaced on screen. The system never auto-escalates, auto-orders, or removes a step from the nurse's existing workflow.
Interaction with clinicians	Nurse enters or reviews vitals/complaint, views the classification and top contributing features, confirms or overrides within 2 actions, documents the reason on override.
Ethics to design translation	Automation-bias risk → the model's output is never framed as a decision; accountability risk → every override is timestamped and reviewable; data-governance risk → the classification is paired with a visible confidence indicator, so low-confidence outputs get more scrutiny, not less.

1.4 Robot Design MVP

This canvas is titled "Robot Design MVP" in the template, and several fields (Robot's role, Personality) are robot-specific. Adapted equivalents for a screen interface are given, per the tutorial's instruction to edit non-robot fields as necessary.

Where and when At the ED triage desk, throughout every shift (day and night); does not change by time of day, though the worst-case design target is hour 11 of a night shift.	**System's role (adapted from "Robot's role")** A decision-support prompt, not an authority — closer to a second pair of eyes than a teacher or an assistant that acts independently.
Draw a picture A desk monitor/tablet showing a triage queue with colour-coded, shape-differentiated patient cards; a single always-visible override control.	**Personality (adapted — N/A for a pure screen interface)** Not applicable; the interface has no persona, voice, or emotional affect. Flagged here rather than filled in, per the template's own "you don't need to fill each one" guidance.
	Context-based behaviour Alert threshold and display format do not change by context; the only adaptive behaviour is switching to a low-confidence flag when the model's certainty falls below a set threshold.
	Connection to systems MVP: none — manual entry only, no EHR pull. A future phase would connect to the Mercer EHR API, with manual entry retained as a fallback.

1.5 Environment

Where The ED triage desk — a small, shared, cluttered physical station. Does not change location.	**User(s)** ED triage nurse.
When	**Secondary user(s)**

Every shift, 24/7; does not change, though conditions (lighting, fatigue, noise) vary by time of day.	Shift supervisor, reviewing override logs; Clinical IT Lead, maintaining the system.
Data collection Manually entered vitals/complaint data at the point of triage; no passive environmental sensing. Stored in the same audit log as override events.	**Simultaneous users** One nurse per desk instance; the interface is not designed for concurrent multi-user input.

1.6 Form

This canvas's fields (Appearance, Character of movement, Voice & sounds, Mobility) are written for a physical robot. Equivalent screen-interface answers are given, with fields marked N/A where a robot-specific question genuinely does not apply.

Draw a picture Triage queue interface. See above.	**Appearance (adapted)** Not applicable in the machine↔lifelike sense; the interface's visual register is clinical and instrumental (colour + icon + numeral), not humanoid or characterful.
	Size Fixed desk monitor or tablet, approx. 10–13".
	Character of movement (N/A) Not applicable — the system has no physical movement.
Voice & sounds No voice output in the MVP; ambient ED noise regularly exceeds 70dB, making audio unreliable as a primary channel. Visual alerting only.	**Mobility (N/A)** Not applicable — the interface is fixed to the triage desk.

2.0. Co-Design Canvas

Setting B: Observation Unit (HRI)

2.1 Problem Space

What problem are you solving?

User			
Who are the users? Are there supporting users?	Group(s)	Characteristics	Needs
Primary Users	Observation Unit nurses monitoring patients post-triage; mixed shift patterns.	Clinically trained but not robotics-trained; manage multiple patients simultaneously; expect predictable, fail-safe behaviour.	Reliable vitals capture without constant supervision; an unambiguous signal on sensor failure.
Secondary Users	Patients in the Observation Unit (co-present, non-expert); porters; visiting family.	Unwell, possibly anxious, may have limited mobility; not clinically trained; did not necessarily choose robotic assessment.	Physical safety and predictable behaviour; clear communication of what the system does and why it is near them.

Goal(s)		
What do the primary and secondary users want to accomplish?	Short-Term	Long-Term
Primary Users	Continuous vitals monitoring without adding nursing workload.	Earlier detection of patient deterioration between nurse rounds.
Secondary Users	Not feeling alarmed or surveilled by the system's presence.	A trusted, well-understood presence in the ward — not an object of suspicion.

2.2 Ethical Considerations

Informed Consent		
	Problem	**Solution**
User	The patient may not know a robotic system is involved in their ongoing assessment, or what it does and does not do.	A visible, plain-language status display at the bedside states what data is being collected and why.
Robot / System	The system has no built-in mechanism to confirm the patient has seen or understood that display.	Consent messaging is added to the Observation Unit admission process, not left to the kiosk alone to communicate.

Physical Autonomy		
	Problem	**Solution**
User	The robot operates close to a patient who may have limited ability to move away.	A hard proximity limit, enforced independently of the ML pipeline, prevents the system from closing distance beyond a safe threshold.
Robot / System	A software fault could in principle instruct behaviour that conflicts with the hardware safety limit.	The proximity sensor is safety-rated and wired to cut power to any actuation, overriding software state entirely.

Cultural Context / Trust		
	Problem	**Solution**
User	In a Caribbean ED, robotic systems may be perceived very differently depending on community familiarity with the technology.	The MVP pilot includes a structured feedback pass with Mercer nursing staff and a small patient sample before wider rollout.

Robot / System	An overly human-like appearance or movement could increase distrust (uncanny valley) rather than reduce it.	The system is deliberately designed machine-like rather than humanoid, avoiding the uncanny-valley risk (see Form, §2.6).

2.3 Guidelines

As with Setting A, Canvas 03's literal field labels were not directly visible in the available materials; the structure below follows the tutorial's description.

Appropriate interaction type	Passive monitoring and vitals capture. The system does not physically intervene, move the patient, or give verbal clinical recommendations to the patient.
Interaction with patients/staff/porters	The robot's status (active/idle/fault) is visible at all times; any escalation is routed to a human nurse, never delivered by the robot as clinical advice.
Ethics → design translation	Consent risk → a visible bedside status display; physical-autonomy risk → a hardware-enforced proximity limit independent of software; cultural-context risk → a structured feedback pass with staff and patients before wider rollout.

2.4 Robot Design MVP

Where and when Beside the existing vitals station in the Observation Unit, it operates continuously across all shifts.	**Robot's role** A monitoring instrument, not a caregiver or companion — closer to an always-on vitals sensor than a helper or assistant with independent judgement.
Draw a picture A stationary console with a status display and sensor array.	**Personality** Deliberately minimal — no simulated emotional states or needs, to avoid the trust/uncanny-valley risk and to keep expectations of the system realistic.
	Context-based behaviour

	On sensor failure, switches to a manual-entry fallback and alerts the nurse rather than continuing on degraded data.
	Connection to systems Streams vitals to the triage classification model; logs faults and overrides to the same audit store as Setting A, so the Board sees one consistent record across both settings.

2.5 Environment (Canvas 05)

Where Fixed position beside the Observation Unit vitals station; does not change location.	**User(s)** Observation Unit nurses (operators).
When Continuously, across all shifts; ambient conditions (noise, humidity, power stability) vary.	**Secondary user(s)** Patients and porters in physical proximity — not operators, but affected by the system's presence and behaviour.
Simultaneous users One patient monitored per unit in the MVP. Trade-off per the template: covering multiple beds from a single unit would require a materially more sophisticated (mobile, multi-sensor) system — explicitly out of scope.	**Data collection** Continuous vitals stream from onboard sensors; stored alongside fault/override events. No patient-identifying display visible to other patients or visitors.

2.6 Form

Draw a picture Kiosk front view + top-down floor placement. See above	**Appearance** Machine-like rather than humanoid or lifelike — positioned toward the "abstract/machine-like" end of the template's scale, not "humanoid". This is a deliberate trade-off: the template notes lifelike robots

	are expected to be more sophisticated in features, which the MVP does not attempt.
Voice & sounds No gendered or characterful voice. Audio is limited to a supplementary fault-alert tone, since ambient noise above 70dB makes voice unreliable as a primary channel; the same tone is used consistently so it is learnable rather than novel each time.	**Size** Smaller than human-size — a stationary console footprint, not a full-body form.
Mobility Does not move across space. Stationary, fixed-base — mobility is explicitly out of scope for the MVP (see 2.4).	**Character of movement** Machine-like, not hybrid or lifelike — the kiosk does not move once positioned; any motion is limited to sensor articulation, not locomotion.

3.0. System Requirements Document

Requirements are written as testable constraints, not feature descriptions. This section is a CariSurg-specific deliverable and is not part of the SoRoCo canvas template.

3.1 Setting A — ED Triage Desk (HCI)

Functional

1. The interface must display the triage classification (1–5) within 1.5 seconds of input submission.
2. The interface must display the top three model features contributing to the classification, in plain clinical language.
3. The system must provide a nurse override mechanism that requires selecting a reason and logs the override with a timestamp.

Non-Functional

1. The interface must remain legible under fluorescent clinical lighting at up to 1 metre (minimum 16pt font, WCAG AA contrast).
2. The system must remain fully usable (read/write) during a generator switchover, with no data loss.
3. Colour-coded alerts must include a secondary non-colour differentiator (icon or shape) for accessibility.

Integration

1. MVP: manual entry only — no EHR dependency at launch.
2. Future phase: the system must pull from the Mercer EHR API via a specified protocol, falling back to manual entry within 2 user actions if the EHR connection is unavailable.
3. All override events must be logged to a store reviewable by the shift supervisor without requiring developer access.

3.2 Setting B — Observation Unit (HRI)

Functional

1. The system must stream vitals from onboard sensors to the classification model continuously while the patient is within range.
2. On sensor failure, the system must switch to a manual-entry fallback within 5 seconds and alert the assigned nurse.
3. The system must never emit a classification derived from partial or degraded sensor data without a visible low-confidence flag.

Non-Functional

1. The system must maintain a minimum patient safety distance enforced by a proximity sensor independent of the ML software stack.
2. The system must degrade to a stationary, non-alerting idle state on power loss, with an immediate alert sent to the nursing station.

3. Status (active/idle/fault) must be visible to nursing staff at a glance from a normal ward working distance.

Integration

1. The system must not display patient-identifying classification data on any screen visible to other patients or visitors.
2. Fault and override events must be logged identically in structure to Setting A's log, so both settings feed one auditable record for the Review Board.

4.0. Safety Considerations

Dr. Reyes's first question will be: "What happens when it fails?" These are documented before he asks.

4.1 HCI Safety Considerations — Setting A

4.1.1. Alarm Fatigue

Concern	Excessive or low-value alerts cause nurses to start ignoring them, reducing the system to background noise.
Context	Mercer ED nurses already manage monitor alerts, medication reminders, and call bells during every shift.
Mitigation	Alert only above a defined confidence threshold; visually distinguish high-confidence urgent flags from low-confidence prompts.
Residual Risk	A threshold calibrated on the Yale EMMLC population (not Mercer's own patients) may be mis-calibrated for the local ESI-1 base rate.

4.1.2. Display Legibility Under Stress

Concern	Interface elements legible in a calm testing environment become illegible under clinical stress, variable lighting, and time pressure.

Context	Triage nurses read the interface while simultaneously managing the patient — it must be readable at a glance, not after several seconds of focus.
Mitigation	Minimum 16pt font for primary information; WCAG AA colour contrast; critical information placed in the top-left quadrant.
Residual Risk	Individual variation in visual acuity and stress response means no single static design works for every nurse in every condition.

4.1.3. Automation Bias / Override Erosion

Concern	Nurses may begin to accept the model's classification by default rather than exercising independent clinical judgement, especially under fatigue.
Context	12-hour shifts, high patient volume (up to ~40 per shift), and a model that is right most of the time create conditions for over-reliance to develop gradually.
Mitigation	The interface never presents the classification as a decision — only as a flag; override requires an active, logged action; override rate is tracked as an MVP success metric.
Residual Risk	Even a logged override mechanism cannot fully prevent a nurse from rubber-stamping the model's output under sufficient time pressure.

4.2 HRI Safety Considerations — Setting B

4.2.1. Sensor Failure / Graceful Degradation

Concern	If onboard sensors fail, the system may produce an incorrect classification or stop functioning mid-assessment.
Context	Caribbean clinical infrastructure includes power fluctuation and humidity-related equipment failure — sensor failure here is foreseeable, not an edge case.
Mitigation	On sensor failure: switch to manual-entry fallback, alert the nurse, log the event. The system never produces a classification on partial sensor data without flagging the gap.
Residual Risk	A nurse mid-shift who receives a manual-fallback request may not have time to respond immediately — resulting in a gap in monitoring rather than a wrong classification.

4.2.2. Proximity / Physical Safety

Concern	A stationary system operating close to a patient could obstruct care, alarm a patient or visitor, or fail to maintain a safe distance if software state is inconsistent.

Context	The Observation Unit kiosk sits beside a vitals station in a shared, sometimes crowded physical space, near patients who may have limited mobility.
Mitigation	A safety-rated proximity sensor enforces a minimum distance independently of the ML software stack, so a software fault cannot compromise physical safety.
Residual Risk	Hardware-level safety does not address patient anxiety or discomfort at having a robotic system nearby — this is an experience risk, not just a physical one.

4.2.3. Voice / Audio Reliability in Noisy Conditions

Concern	ED and Observation Unit ambient noise regularly exceeds 70dB, causing high error rates in any voice-based interaction and reducing the reliability of audio-only alerts.
Context	Generator noise, HVAC systems, and patient volume are standard operating conditions in the Mercer environment, not unusual disturbances.
Mitigation	Audio is supplementary only, never the sole channel for a fault alert; visual status display is the primary channel; no voice-command interaction in the MVP.
Residual Risk	Nurses occupied elsewhere in the ward may still miss a visual-only fault signal if it is not paired with an escalation path.

4.3. Failure Modes — What Happens When Each Setting Breaks

Failure Event	Setting A — HCI	Setting B — HRI
Power Loss	Screen goes dark; nurse defaults to the manual (paper/EHR) triage protocol.	Kiosk freezes/idles; patient assessment pauses; immediate alert sent to the nursing station.
Connectivity Loss	EHR pull fails (future phase); fallback to manual entry; alert displayed on screen.	Device stream drops; graceful degradation activates; manual-entry fallback within 5 seconds.
Sensor Failure	N/A for the MVP (manual-entry only, no device stream); revisit if a future device stream is added.	Proximity or vitals sensors offline; kiosk moves to stationary/safe mode; manual override to a fully stationary state.
User Error	Nurse enters an incorrect value; model outputs a misleading classification; override mechanism available and logged.	Patient or visitor enters a restricted proximity zone; safety protocol activates independently of software state.
Low Model Confidence	Interface displays a low-confidence flag alongside the classification; nurse is prompted to review manually.	System prompts a manual review; low-confidence classification is not surfaced on any patient-facing display.

For each failure mode: (a) what the system does, (b) what the human does, (c) how the event is logged — all three are captured identically across both settings so the Review Board sees one consistent audit trail.

Inputs, Outputs & Human Action

Comparison

Dimension	**Setting A — ED Triage Desk (HCI)**	**Setting B — Observation Unit (HRI)**
Who is the user?	ED triage nurse — trained clinical staff.	Mixed: observation nurses (operators), patients and porters (in proximity, non-operators).
What data does the model receive?	Manual entry in the MVP (age, presenting complaint, vitals); EHR pull deferred to a future phase.	Device stream from onboard vitals sensors; manual-entry fallback within 2 user actions if the stream drops.
What does the model emit?	Colour-coded alert (red/amber/green) + numeral (1–5) + top three contributing features; secondary icon/shape differentiator.	Visual status display (active/idle/fault) plus a classification routed only to the nurse's screen, never patient-facing.
What does the human do next?	Review, confirm or override (with a logged reason), document the decision.	Review the flagged status, respond to the vitals reading or fallback prompt, document the encounter.

Who to Ask?

Question About	**Contact**
Model Choice	Josiah-John Green
Dataset Access	Martina Griffith (Clinical IT Lead)
Clinical Validity of ESI Predictions/Triage Workflows	Dr Marcus Reyes (Consultant EP) or Dr De Freitas (Clinical Sponsor)
Programme Process/Submission Logistics	Carisurg Tutor Team, in ‘#ask-a-tutor’

Appendices

Supplementary Links

1. The **GitHub link** can be found here.
2. The **Figma link** can be found here.
3. The **TinkerCAD** link can be found here.
4. The **Loom** video link can be found here.

Glossary

A reference index for clinical, technical, and research terminology used throughout this proposal. Intended for any reviewer — clinical or non-clinical — who needs a quick definition without breaking flow.

Clinical & Vital Sign Terms

Term	**Meaning**
ED	Emergency Department
ESI	Emergency Severity Index — the 5-level triage scale used at Mercer General (1 = immediate, 5 = non-urgent)
SBP	Systolic Blood Pressure — the peak pressure during a heartbeat (top number in a BP reading)
DBP	Diastolic Blood Pressure — the pressure between heartbeats (bottom number in a BP reading)
MAP	Mean Arterial Pressure — average pressure driving blood to organs across a cardiac cycle; calculated as (SBP + 2×DBP) / 3
GCS	Glasgow Coma Scale — score from 3–15, measuring level of consciousness
RR	Respiratory Rate — breaths per minute

FiO2	Fraction of Inspired Oxygen — the percentage of oxygen a patient is receiving (room air = 21%)
SpO2	Oxygen Saturation — percentage of haemoglobin carrying oxygen, measured via pulse oximeter
SBAR	Situation, Background, Assessment, Recommendation — a structured format for clinical handover communication

Healthcare Systems & Infrastructure Terms

Term	**Meaning**
EHR	Electronic Health Record — digital patient record system; Mercer currently has none for triage, hence the Phase 1 proposal
LMIC	Low- and Middle-Income Country — a World Bank income classification used in global health literature
SIDS	Small Island Developing State — a UN classification for small island nations facing shared infrastructure constraints
Boarding time	The gap between a disposition decision (e.g. “admit”) and the patient physically leaving the ED, usually due to no ward bed being available

AI / Machine Learning & Risk Terms

Term	Meaning
AI	Artificial Intelligence
ML	Machine Learning
AUROC / AUC	Area Under the Receiver Operating Characteristic curve — a model performance metric from 0.5 to 1.0; higher is better
XGBoost	Extreme Gradient Boosting — a machine learning algorithm using an ensemble of decision trees, common for structured/tabular data
Rule-based classifier	A decision-making system using explicit if/then logic rather than a trained statistical model
PPV	Positive Predictive Value — the proportion of positive alerts that turn out to be true positives. Low PPV drives alert fatigue.
Automation bias	The tendency for a human user to over-trust or defer to an automated system's output, even when contradicted by their own judgment
Distribution shift	When a model's performance degrades because the data it encounters in deployment differs statistically from the data it was trained on
Human-in-the-loop	A system design principle requiring a human to review, confirm, or be able to override an AI-generated output before it takes effect

Research & Documentation Terms

Term	Meaning
DOI	Digital Object Identifier — a permanent link format for academic papers
APA	American Psychological Association — the citation style used throughout this proposal (7th edition)
Scoping review	A literature review mapping the breadth of existing research, without necessarily assessing study quality in depth
Systematic review	A more rigorous literature review following a defined search and inclusion protocol